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MILITARY SPECIFICATION

TANKS, METAL FINISHING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers metal finishing tanks and related equipment used in electroplating, anodizing, nonelectrolytic plating and coating processes with equipment and accessories specified herein.

1.2 Classification. The tanks shall be of the following styles, types, and classes. The style, type and class tank to be furnished shall be as specified (see 6.2.1).

Style A	- Tank with anode and cathode rods
Style B	- Tank without anode and cathode rods
Type I	- Polyvinyl chloride lined tank
Type II	- Rubber lined tank
Type III	- Neoprene lined tank
Type IV	- Lead lined tank
Type V	- Stainless steel tank
Type VI	- Steel tank
Type VII	- Copolyvinyl tank
Type VIII	- Polyvinyl tank
Type IX	- Polyester lined tank
Type X	- Polypropylene lined tank
Type XI	- Polyester tank
Class 1	- Electrically heated tank
Class 2	- Steam or hot water heated tank
Class 3	- Gas heated tank
Class 4	- Unheated tank

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the non-Government documents which are current on the date of the solicitation.

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA ICS 1 - Industrial Control and Systems.

ANSI/NEMA RI-8 - Silicon Rectifiers.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

Welding Handbook, Fifth Edition

(Application for copies should be addressed to United Engineering Center, 345 East 45th Street, New York, NY 10017.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The tank shall be new and one of the manufacturer's current models capable of operations in accordance with the requirements herein. The tank shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. Design and construction of the tanks shall provide leakproof safe operation. Design characteristics shall be in accordance with 3.2.1 through 3.2.3.4 for each style, type, and class tank specified. Unless otherwise

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specified (see 6.2.1), all required service piping and drains shall be located on the length dimensions of the tanks.

3.2.1 Styles.

3.2.1.1 Style A. Tanks shall be provided with mounted anode and cathode rods and the rods shall be constructed of solid copper. The rod diameter, round or square, number of each type rod required, and mounting arrangements shall be as specified (see 6.2.1), (rod diameter should be based on a minimum of 1 square inch of copper per 1000 amps). Rod mounting supports shall be of the sliding saddle type for maximum versatility of rod spacing.

3.2.1.2 Style B. Tanks shall be provided without anode and cathode rods.

3.2.2 Types. Tanks that require lining shall be sandblasted to white metal inside and over the rim. All welded and uneven surfaces shall be ground smooth with all splatter and flux removed. Each lining shall completely cover the inside of the tank and extend over and under the rim.

3.2.2.1 Type I. Tanks shall be steel construction and lined with plasticized polyvinyl chloride (PVC) material. The lining shall be koroseal, or amerplate type material or equal, and shall be not less than 3/32-inch thick on all sides and over the rim, and not less than 3/16-inch thick on the bottom.

3.2.2.2 Type II. Tanks shall be steel construction and lined with natural rubber. The lining shall be not less than 3/16 inch in thickness. The lining shall be applied in unvulcanized sheet over suitable adhesives and vulcanized to the tank surface under heat and pressure.

3.2.2.3 Type III. Tanks shall be steel construction and lined with precured high temperature neoprene (synthetic-rubber) material. The lining shall be not less than 3/32 inch in thickness on all sides and 3/16-inch thick on the bottom.

3.2.2.4 Type IV. Tanks shall be steel construction and lined with 8 pound per square foot (1/8 minimum thickness) lead. Unless otherwise specified (see 6.2.1), the lining shall be 6 percent antimony alloyed lead.

3.2.2.5 Type V. Tanks shall be unlined and fabricated of stainless steel of the type specified (see 6.2.1). Tanks up to 5 feet in length shall be fabricated of material not thinner than 14 gauge, and from 5 to 10 feet not thinner than 10 gauge, and tanks greater than 10 feet in length shall be not less than 3/16 inch in thickness. Each tank shall have stainless steel girths, and side braces as specified for steel tanks.

3.2.2.6 Type VI. Tanks shall be unlined and fabricated of mild steel plate in accordance with 3.2.4.

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3.2.2.7 Type VII. Tanks shall be fabricated from copolyvinyl-chloride (CPVC) plastic material not less than 1/4-inch thickness. Tanks of capacity from 16 to 30 gallons shall have one polyvinyl-chloride (PVC) angle girth not less than 1/4 inch by 2 inches by 3 inches located at the top outside, and cemented to the sidewalls. Tanks of capacity from 30 to 60 gallons shall have two PVC angle girths not less than 1/4 inch by 2 inches by 2 inches, one located at the top and one located 1/3 up from the bottom of the tank, cemented to the sidewalls. Tanks of capacity greater than 60 gallons shall be not less than 3/8 inch in thickness and have not less than two PVC coated steel, stainless steel, or PVC angle girths, one located at the top and one located 1/3 up from bottom of the tank secured to the sidewalls. Girths shall be of adequate number and size to prevent sidewall deflection or building as specified herein.

3.2.2.8 Type VIII. Tanks shall be fabricated from PVC plastic material of the thickness and girded as specified for type VII tanks.

3.2.2.9 Type IX. Tanks shall be steel constructed and lined with glass fiber or flake reinforced polyester resin material not less than 0.080 inch thickness.

3.2.2.10 Type X. Tanks shall be steel constructed with a liner of polypropylene material not less than 1/4-inch thickness. Adhesion to tank side walls is not necessary. The liner shall cover the inside of the tank and tuck under the tank rim.

3.2.2.11 Type XI. Tanks shall be of laminated glass fiber reinforced polyester resin construction. The sidewalls and bottom of the tank shall be not less than 1/4-inch thickness, and the sidewalls shall be girded as specified for type VII tanks.

3.2.3 Classes. The class of tank shall be determined by the heating method. The design and arrangement of the heating system for each class tank shall be sufficient to maintain bath temperature with deviation of not greater than $\pm 10^{\circ}$F. Each heated tank shall be provided with an automatic temperature control system complete with temperature indicating thermometer. Unless otherwise specified (see 6.2.1), the temperature controls for electric heated tanks shall be of the shockproof, variable thermostat type; and the controls for steam-heated tanks shall be of the direct acting indicating type. The temperature control system for each class tank shall be capable of controlling the temperature with a deviation of not greater than $\pm 5^{\circ}$F from any preset temperature within the range of the system. The temperature control range for each steam and electric heated tank shall be from not greater than 90° F through not less than 225° F. All valves, bulbs, traps, and parts that come in contact with the bath solution shall be stainless steel or compatible with the tank lining. All heaters shall be provided with hanger supports and designed for locating not less than 2 inches off the bottom of the tank.

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3.2.3.1 Class 1. Tanks shall be heated by sheathed immersion type electric heaters of the hang over the side design. The heater sheath materials shall be as specified (see 6.2.1), table I and 6.5). All heaters shall have adequate shields. The KW electrical input or the heating rate required of each tank shall be as specified (see 6.2.1).

3.2.3.2 Class 2. Tanks shall be steam or hot water heated. Unless otherwise specified, class 2 tanks shall be heated by 16-gauge, Type 304 stainless steel embossed sheet metal coils. When specified (see 6.2.1), class 2 tanks shall be heated by either mild steel embossed coil, Type 316 embossed stainless steel coil, steam lead coil, teflon tube, titanium embossed sheet coil, or coil steel pipe as specified. Heating systems for electrified tanks shall have nonconductor insulating sections for both inlet and outlet connections. Heating requirements shall be based on a steam pressure of 20 psi or 260° F hot water and the tank filled to not less than 80 percent of capacity with water at 80° F. The heat up time and maximum temperature shall be as specified (see 6.2.1).

3.2.3.3 Class 3. Tanks shall be heated by either gas burners under the tank, heater tubes constructed within the tank sidewalls, or immersion-type gas heaters as specified (see 6.2.1). Unless otherwise specified, the heating system shall operate from natural gas having a nominal heat content of 1000 BTU/cu. ft. and a supply pressure of 6 through 8 inches water column. When required, the heaters shall be designed and equipped to operate on either a butane or propane fuel supply as specified (see 6.2.1). After initially starting, the heating system shall operate automatically and shall maintain a bath uniform temperature within ±10° F of any preset temperature. The operating temperature range of the tank shall be as specified (see 6.2.1). For temperatures above 200° F, all tank sidewalls shall have not less than 2 inches of insulation with 10-gauge steel covering the outside, and for temperatures above 230° F, all tank sidewalls shall have not less than 2-inch hot-air chamber between the insulation and inner wall. Heating requirements shall be based on the tank filled to 80 percent of capacity with water at 80° F and the heat-up time to the specified operating temperature shall be as specified (see 6.2.1).

3.2.3.4 Class 4. The class 4 tank shall be unheated.

3.2.4 Material. All material used in the fabrication of the tanks and related equipment shall be of the quality necessary to produce tanks and equipment to meet the requirements described herein. Except as specified for type V tanks (see 3.2.2.5), all steel tanks with volumes up to 18 cubic feet and up to 36 inches in depth shall be fabricated from material not thinner than 12 gauge and up to 18 cubic feet, over 36 inches in depth, not thinner than 10 gauge . Tanks from 18 to 60 cubic feet up to 36 inches in depth shall be fabricated from material not thinner than 10 gauge and from 18 to 60 cubic feet over 36 inches in depth from material not thinner than 3/16 inches. All tanks over 60 cubic feet in volume shall be fabricated from material not less than 1/4 inches in thickness. The material used in fabrication of the tanks shall be mild steel unless specified otherwise.

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All plate material shall be flat and free from buckling. All material thickness, rims, girths and brace requirements specified herein are minimum required. Each tank shall be designed, constructed and reinforced as necessary to limit rim deflection to not more than 1/8 inch when filled with water. All fastening devices, pipe and other parts that come in contact with plating solutions shall be stainless steel, rigid plastic or constructed of material to prevent corrosion and abrasion.

3.2.5 Reclaimed materials. The tank may contain reclaimed materials provided such materials will not jeopardize the tank's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the tank.

3.2.6 Energy efficiency. The tank and its components that directly consume energy in normal operation shall be designed and constructed for the energy efficiency as governed by the developments available within the industry.

3.2.7 Mercury restriction. The tank shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.8 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the tank.

3.2.9 Environmental protection. The tank shall be so designed and constructed that under the operating, service, transportation, and storage conditions described herein, the tank shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at the point of installation (see 6.4).

3.2.10 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the tank. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions sizes, and capacities are given in US units. The measurement may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.11 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.12 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.2.13 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the tank that present safety

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hazards. The safety devices shall not interfere with the operation of the tank. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All tank parts, components, mechanisms, and assemblies furnished on the tank, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the tank itself. Additional safety and health requirements shall be as specified (see 6.2.1).

3.3 Construction. The tank shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the tank to its maximum rating and capacity without distortion.

3.3.1 Tank rim. Each tank shall be reinforced around the top outside with an angle or channel iron (rim) girth. The girth may be formed by bending a 3-inch section of the top plate and a 1-inch vertical flange to anchor the insulation retaining sheet metal on the outside. If angle or channel iron is used, the girth for tanks with volumes up to 18 cubic feet shall be not less than 3-inch angle or channel iron. The thickness of the girth shall be equal to or greater than the wall thickness. The upper edge of the tank and girth shall be jointed with a continuous arc-weld and the lower edge of the girth shall be spotted with welds from 1-1/2 to 2 inches long, placed every 6 to 8 inches apart.

3.3.2 Side braces. Tanks fabricated from steel over 6 feet in length and 3-1/2 feet deep shall be provided with a horizontal angle girth of not less than 3-inch angle iron. The horizontal girth shall be located 1/3 the way up from the bottom of the tank and welded as specified for the top girth. Tanks over 10 feet long and 4-1/2 feet deep shall be provided with vertical gussets located on approximately 3-1/2 foot centers in addition to the horizontal girth. When necessary to prevent any appreciable bulging, tanks shall be vertically gussetted with structural steel members welded at right angles to the side of the tank.

3.3.3 Tank base supports. When specified (see 6.2.1), the bottom of each steel tank shall be provided with base supports of structural steel I-beam or similar nonporous material. The supports shall raise the tank not less than 4 inches. Support hall be located crosswise and not greater than 40 inches apart or as required to accommodate a machine base or foundation piers.

3.3.4 Drains. When specified, tanks shall be provided with drain outlets. The drain shall be located on the service piping side, at the bottom, and near one corner of the tank. The drain size shall be 1-1/2-inch pipe size (IPS) for tanks up to 18 cubic feet, 2 inches IPS for tanks from 18 to 40 cubic feet, and 3 inches IPS for tanks with volumes over 40 cubic feet. All drains for lined tanks shall be provided with flanged outlets in accordance with 3.3.4.1. The style, type, and class of each tank to be provided with drains shall be specified (see 6.2.1).

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3.3.4.1 Flanged drain outlets. All drains for lined tanks shall be provided welded neck flanges or have flanged nipple type outlets. The nipple shall be 4 to 6 inches long, of double strength pipe, and welded both inside and outside to the tank. The flanged shall be screwed securely to the nipple and welded, both inside and outside of the flange. The lining shall extend through the nipple or flange neck and cover the flange surface to form a gasket for the mating flange. When specified (see 6.2.1), mating double flange type valves with a blind flange shall be provided for each drain outlet. The valves shall be of the lined diaphragm type with the valve lining and diaphragm material compatible with the lining of the tank.

3.3.5 Overflow dam. When required, tanks shall be provided with overflow dams. Overflow drains for lined tanks shall be of the flanged type. The type of dam, location of the dam in the tank, and the style, type, and class of each tank to be equipped with dam shall be as specified (see 6.2.1).

3.3.6 Sumps. When required, tanks shall be designed with drain sumps. Drains outlets shall be located on the side and at the bottom of the sump. The style, type, and class of each tank to be provided with sump drain shall be as specified (see 6.2.1).

3.3.7 Welds. Steel tanks shall be constructed with continuous electric welded seams. Steel tanks greater than 60 inches in any dimensions shall be constructed with continuous double electric welded seams. Welding electrodes shall conform to American Welding Society specifications, as specified in AWS Handbook, Section 5, for the parent metal to be welded.

3.3.8 Brazing or soldering. Brazing or soldering shall be employed only where those operations are required in the original design. These operations shall not be employed as repair measures for defective parts.

3.3.9 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.10 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.11 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

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3.3.11 Painting. The outside of each metal tank (except stainless steel tanks) shall be painted with acid- and alkali-resistant paint or coated with a plastic material.

3.3.12 Threads. All threaded parts used on the tank and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.4 Performance. Tanks shall be capable of continuous operation in chemical plating and anodizing processes, without damage to the lining, tanks or heater jackets using the specified chemical solution concentration at the temperature listed for each type tank in table I. Concentrations specified are percent by volume of the usual market concentrations in water.

3.4.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.5 Components. The tank mechanisms and accessories shall be furnished complete with the required electric motors, starters, contactors, controls, relays, transformers, rectifiers, signal lights, and switches, prewired where possible for operation upon connection to source of power. All power actuated, manually adjusted, and automatic controls shall operate on nominal 120 volts or less power supply. All electrical equipment shall conform to NEMA ICS-1 and enclosures shall be NEMA Type 12.

3.5.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Motors rated in excess of 1/2 horsepower shall be dual voltages, 230/460 volt, and operate on the electrical characteristics specified for rectifiers. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure.

3.5.2 Rectifiers. Rectifiers shall be furnished when specified (see 6.2.1). The rectifiers shall be of the silicon diode hermetically sealed type, and shall conform to NEMA RI-8. The rectifiers shall be either air cooled or water cooled as specified (see 6.2.1). Each rectifier shall be capable of operating continuously in ambient temperature of 104° F. Transformer insulation shall be NEMA Class B or better. Each rectifier shall be provided with a DC voltmeter and DC ammeter with a calibrated accuracy of ±2 percent of full scale reading. The input voltage, cycle, phase, and the output DC voltage range and amperage requirements shall be as specified (see 6.2.1). Controls shall be provided in accordance with 3.5.2.1.

3.5.2.1 Rectifier controls. The voltage control system for each rectifier shall be as specified (see 6.2.1), and shall be selected from the systems listed in table II. When required, remote control shall be provided with any system specified, and automatic and remote control requirements shall be as specified (see 6.2.1).

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TABLE I. Heater sheath material and solution temperature for specified chemcial solutions.

	Heater Sheath Material	Types VII and X	Types I and VIII	Type II	Type III	Types IX and XI
Acids						
Hydrochloric 25%	Q	180	140	180	X	180
Sulfuric 50%	Q	180	130	150	150	180
Sulfuric 10%	Q	180	140	150	150	180
Nitric 50%	Q	X	70	X	X	70
Nitric 10%	Q	180	140	X	X	150
Chromic 20%	Q	180	130	X	X	150
Fluoboric 40%	G	180	140	150	150	180
Hydrofluoric 50%	G	180	70	150	X	---
Acetic Dilute 60%	Q	180	X	150	X	180
Phosphoric 80%	Q	180	140	150	200	180
Acid Plating						
Sulfate	Q	180	140	150	200	180
Sulfate Chloride	Q	180	140	150	200	180
Chloride	Q	180	140	150	X	180
Fluoborate	G	180	140	150	150	180
Sulfamate	Q	180	140	150	150	180
Alkalies						
Ammonia Liquid	S/S	X	X	150	200	X
Caustic Soda 35%	S/S	130	90	150	150	---
Caustic Soda 10%	S/S	150	140	150	200	---
Trisodium Phosphate	S/S	180	140	150	200	180
Soad Ash	S/S	180	140	150	200	180
Silicates	---	180	140	150	200	180
Cyanides	S/S	180	140	150	200	---
Alkali Plating						
Cyanide	S/S	180	140	150	200	---
Stannate	S/S	180	140	150	200	---
Pyrophosphate	S/S	180	140	150	200	---
Amine	S/S	180	140	150	200	---

NOTES:
X = Not recommended (see 6.7).
Q = Quartz.
G = Graphite.
S/S = 304 Stainless steel.

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TABLE II. Rectifier controls.

Control System	Method	Control Range	Automatic Controls	Power Factor	Maximum Ripple*
Limited Tap Switch	4 Step	75-100%	No	90%	4.2%
Tap Switch Half Range	22 Step	50-100%	No	90%	4.2%
Tap Switch Full Range	22 Step	20-100%	No	90%	4.2%
Variable Transformer	Stepless	0-100%	Motorized Optional	90%	4.2%
Reactor Primary	Stepless	10-100%	Optional	70-80%	10% at Rated Output Voltage
Reactor Primary	Stepless	0-100%	Optional	70-80%	
Reactor Secondary	Stepless	10-100%	Optional	70-80%	
Reactor Secondary	Stepless	0-100%	Optional	70-80%	

NOTE: *Base on 3-phase rectifier.

3.5.3 Heat exchanger. When specified, tanks shall be provided with a heat exchanger cooling system. The system design requirements, type of system required (either internal or external), the cooling medium required, type, and class of tank to be cooled and the chemical solution concentration the system will be used with shall be as specified (see 6.2.1). The materials used in construction of the cooling system shall be compatible with the solutions specified.

3.5.4 Insulation. When specified, tanks shall be insulated on all four sides with not less than 1 inch of insulation material and covered with sheet metal. The style, type, and class of each tank to be insulated shall be as specified (see 6.2.1). (Class 3 tanks shall be insulated as specified in 3.2.3.3.)

3.6 Sizes. The sizes of the tanks shall be as specified (see 6.2.1). The minimum size of the tanks shall be not less than 24 inches in length by 18 inches in width by 18 inches in depth. Size changes shall increase in increments of 6 inches in length, width, or depth as required up to a maximum of 12 feet in length 5 feet in width and 5 feet in depth. Dimensions specified shall be inside dimensions and shall vary not more than plus or minus 1/4 inch.

3.7 Barrel process. When specified (see 6.2.1), tanks shall be fitted for barrel plating. The style, type, and class of tank to be fitted and barrel performance requirements shall be specified.

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3.8 Standard equipment. Unless otherwise specified, all standard equipment normally furnished by the manufacturer shall be furnished. Additional equipment shall be furnished as specified and fully described (see 6.2.1).

3.9 Optional equipment. The following optional equipment shall be furnished as specified (see 6.2.1). Performance and design requirements shall be as specified for each item or system.

Exhaust ventilation systems
Fume washers
Steam generators
Control valves, flowmeters
Transfer pumps
Immersion portable electric heaters
Heating elements
Temperature controllers
Agitator system (air or rod)
Solution filter systems
Plating, cleaning and chemical supplies
Storage and mixing tanks
Work baskets and racks
Hoist and handling equipment for nonautomated tank lines
Anode baskets and bags
Cathodes and anodes
Extra rods
Plating barrels
Extra-immersion cooling or heating coils
Magnets
Ph control system
Tank spray systems
Testing equipment
Timing devices

Load and unload equipment
Rack storage
conveyor systems
Effluent treatment equipment
Deionized water treatment equipment
Prepiping of tanks
Interconnecting piping for accessories
Chemical recovering equipment
Tank covers

3.10 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.11 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

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3.11.1 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each tank and contain the information listed below. If the tank is a special model, the model designation shall include the model of the basic standard tank and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Type lining
Inside dimension
Operating temperature range, if heated (degree F)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.13 Workmanship. Workmanship of the tank and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not

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authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination or any test shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.8. Failure of the item to pass the examination, the tests, or the inspections shall be cause for rejection.

4.6 Examination. The tank shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Circuit test. Each circuit of the electrical system shall be tested for continuity, short circuits and faulty ground.

4.7.2 Leakage test. Each tank shall be tested for leaks. All weld fabricated tanks shall be cleaned as specified in 3.2.2 and the test performed before the lining or insulation is installed. The testing method shall be of magnetic particle, dye penetrant, X-ray or other commercially accepted test methods of detecting welded devices. If any leaks are detected they shall be repaired and the tank retested.

4.7.3 Operational inspection. After the leakage test specified in 4.6.2 has been completed, controlled temperature tanks shall be connected to the necessary supply sources and tested for heating or cooling capability and temperature uniformity specified for the type of tank under test.

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5.8 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The tanks covered by this specification are intended for use in electroplating, anodizing, nonelectrolytic plating and coating processes.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Style, type, class, and number required of each tank (see 1.2).

c. Size required, length, width, and depth in inches (see 3.6).

d. First article when required (see 3.1)

e. Service piping location, if different (see 3.2).

f. Cathode and anode rod requirements; style A tanks only (see 3.2.1.1).

g. Type of lead lining, if different, type IV tanks only (see 3.2.2.4).

h. Type stainless steel tank material required (see 3.2.2.5) (specify 316, 316L, 304, etc.)

i. Temperature controls, if different (see 3.2.3).

j. Type of electric heater sheaths, and required heating rate or KW input (see 3.2.3.1, table I and 6.5).

k. Type of steam heaters, if different, class 2 tank only (see 3.2.3.2).

l. Heating requirements, class 2 tanks only (specify heating rate) (see 3.2.3.2).

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m. Specify type of gas heating system, type of fuel gas and heating rate (see 3.2.3.3) (class 3 tanks only).

n. Measurement system, if different (see 3.2.10)

o. Specify exceptions and safety and health requirements (see 3.2.13 and 6.4).

p. Tank base supports, when required (see 3.3.3).

q. Drains, when required (see 3.3.4).

r. Double flange type valves, when required (see 3.3.4.1).

s. Overflow dams, when required (see 3.3.5).

t. Sump drains, when required (see 3.3.6).

u. Additional performance features, if required (see 3.4.1).

v. Motor enclosure, if different (see 3.5.1).

w. Rectifiers, when required (see 3.5.2).

x. Rectifiers cooling method required (see 3.5.2).

y. Rectifiers electrical characteristics, specify (see 3.5.2).

z. Type of voltage control system required for each rectifier (see 3.5.2.1).

aa. Rectifier remote control and automatic controls, when required (see 3.5.2.1).

bb. Heat exchanger cooling system, when required (specify all options required) (see 3.5.3).

cc. Tank insulation, when required (see 3.5.4).

dd. Tank designed and fitted for barrel processing, when required (see 3.7).

ee. Standard equipment equipment, if different (see 3.8).

ff. Optional equipment, when required (see 3.9).

gg. Repair parts, when required (see 3.10).

hh. Nameplate, when required (see 3.11.1).

ii. First article inspection, if different (see 4.4).

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jj. Quality conformance inspection, if different (see 4.5).

kk. Preparation for delivery, if different (see 5.1).

6.2.2 Contract data requirements. Required technical data (operating manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Tank lining. Table I may be used as a guide in selecting the type of tank or lining and electric heater (sheath) material recommended for a particular solution.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Subject term (keyword) listing:

Anodizing
Coating
Electroplating
Plating
Tanks

Custodians:
Army - AL
Navy - SH
Air Force - 99

Preparing activity:
DLA - IP

Project (3426-0039)

Review activities:
Air Force - 84
DLA - GS

User activities:
Navy - AS, OS, MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-80144C	Tanks, Metal Finishing

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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